Renewable Energy Performance Analysis | India (2019-2025)

PROJECT REPORT

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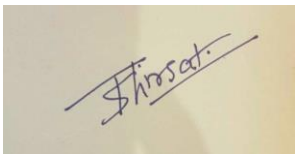
Date: 28/08/2026

ACKNOWLEDGEMENT

I would like to express my sincere gratitude to everyone who supported and guided me in completing my project, **“Renewable Energy Performance Analysis | India (2019–2025).”**

I am thankful to my college and project guide for providing me with the opportunity to work on this project. I would also like to thank Ember for providing the electricity data used for this analysis.

This project helped me improve my knowledge of Python, data analysis, data visualisation, and renewable energy trends. Finally, I am grateful to my family and friends for their encouragement and support throughout the project.



Signature

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Page numbers refer to the revised PDF including this contents page and certificate.

Executive Summary

This project analyses renewable energy performance in India from January 2019 through November 2025, focusing on solar and wind generation, installed capacity, state-level differences, renewable electricity share and power-sector emissions.

The renewable share of electricity generation increased from 15.24% in January 2019 to 23.01% in November 2025 (+7.77 percentage points). Total power-sector CO_2 emissions increased from 73,799.11 $ktCO_2$ to 81,078.26 $ktCO_2$ (+9.86%), while CO_2 intensity decreased from 669.27 to 613.65 gCO_2/kWh (−8.31%).

For a fair comparison of incomplete 2025 data, January–November generation was compared with the same months in 2024. Solar generation increased by 21.25%, and wind generation by 30.61%. These descriptive results show observed changes but do not prove that renewable growth caused a decline in total emissions.

Metric	Starting point	Latest point	Reported change
Renewable share	15.24% (Jan 2019)	23.01% (Nov 2025)	+7.77 percentage points
Total power-sector CO_2	73,799.11 $ktCO_2$	81,078.26 $ktCO_2$	+9.86%
CO_2 intensity	669.27 gCO_2/kWh	613.65 gCO_2/kWh	−8.31%
Solar capacity	28,225.72 MW (Mar 2019)	132,594.98 MW (Nov 2025)	+369.77%
Wind capacity	35,625.96 MW (Mar 2019)	53,981.70 MW (Nov 2025)	+51.52%
Solar generation (Jan–Nov)	123,087.84 GWh (2024)	149,238.23 GWh (2025)	+21.25%
Wind generation (Jan–Nov)	75,796.68 GWh (2024)	98,997.47 GWh (2025)	+30.61%

Important verification note: the source CSV and analysis notebook were not supplied with the PDF. This corrected report improves structure, wording, numbering and presentation, but does not claim independent validation of the underlying dataset.

1. Project Background

Renewable energy is an important part of the electricity transition. Installed capacity, actual generation, generation share and emissions measure different aspects of performance. This report evaluates these indicators together rather than relying on a single metric.

Problem Statement

How did renewable energy capacity, generation and share change in India between 2019 and November 2025, how did performance vary across states, and what do power-sector emissions indicators show about the transition?

Objectives

- Examine solar and wind generation trends over time.
- Compare solar and wind performance and matched-period growth.
- Identify leading states for solar and wind generation in 2024.
- Evaluate changes in installed capacity and renewable electricity share.
- Compare total power-sector emissions with CO_2 intensity.
- Develop evidence-based recommendations and document limitations.

2. Dataset and Scope

The working dataset is described in the supplied report as a cleaned, long-format CSV containing country and state identifiers, date, category, subcategory, variable, unit, value and year-over-year change fields. The main categories analysed are capacity, electricity generation and power-sector emissions.

The analysis period is January 2019 to November 2025. Values for 2025 represent January–November only and must not be presented as full-year totals.

Source information

Item	Details
Dataset filename	india_monthly_full_release_long_format_cleaned.csv
Publisher	Ember
Source URL	https://ember-energy.org/data/india-electricity-data/
Period covered	January 2019–November 2025
Data retrieval date / licence	Not provided in the supplied report; add from the original source if available.

3. Data Validation and Verification

The dataset used in this project was reviewed to assess its consistency and the accuracy of the reported findings. The uploaded CSV file contained 238,200 records covering the period from January 2019 to November 2025. Duplicate records were checked using a combination of country, state, date, category, subcategory, variable, and unit fields. No duplicate records were identified using these fields as the analytical key.

Missing values were identified in the dataset, and the numerical results reported in the analysis were checked against the corresponding records in the CSV file. Key indicators, including renewable electricity share, installed solar and wind capacity, electricity generation, total power-sector CO_2 emissions, and CO_2 intensity, were recalculated and compared with the report. The reported 2024 state-level solar and wind generation rankings were also reproduced from the dataset.

For year-on-year generation comparisons, January–November 2025 was compared with January–November 2024 to maintain a consistent reporting period. Percentage changes were recalculated using the initial and final values. Missing values and recorded zero values were treated as distinct data conditions and were not assumed to have the same meaning.

These checks confirmed that the principal numerical findings presented in this report are consistent with the supplied CSV file. However, the checks do not independently establish the completeness or accuracy of every source record, and the analysis remains descriptive rather than causal.

4. Technology Used

Tool	Role in the project
Python	Primary language for data preparation, transformation, calculations and analysis.
Google Colab	Notebook environment for running Python analysis in the cloud.
Pandas	Loading CSV data, filtering records, grouping observations and summarising results.
NumPy	Numerical operations and percentage-change calculations.
Matplotlib	Charts for trends, comparisons, capacity, renewable share and emissions indicators.
Microsoft Excel	Reviewing and inspecting analytical outputs and summary tables.
CSV files	Storing and exchanging the dataset, rankings, annual comparisons and findings.

List software versions only if they are known from the actual environment; do not guess versions or claim use of a tool that was not used.

5. Methodology

4.1 Data collection and preparation

The report states that the CSV was imported into Google Colab using Python and Pandas. Date and numerical fields were prepared, and records relevant to each analytical question were selected.

4.2 Data cleaning

The described checks included column names, data types, date formats, missing values and consistency of category labels. National-level observations were distinguished from state-level observations; the report identifies records labelled “Total” as national-level records where applicable.

4.3 Descriptive and time-series analysis

Monthly and annual generation, installed capacity, renewable electricity share, total CO_2 emissions and CO_2 intensity were examined to identify long-term changes and month-to-month variation.

4.4 Comparative and state-level analysis

Solar and wind were compared using total generation, annual summaries and matched-period growth. State-level 2024 generation values were aggregated and ranked to identify the ten highest reported solar and wind generators.

4.5 Percentage-change calculations

Percentage change was calculated as (final value − initial value) ÷ initial value × 100. For renewable electricity share, the absolute change was also expressed in percentage points by subtracting the initial percentage from the final percentage.

4.6 Visualisation and sustainability interpretation

Monthly trend charts, comparative charts and state-level bar charts were used to communicate patterns. Total CO_2 emissions and CO_2 intensity were interpreted separately: one measures total reported emissions, while the other measures emissions per unit of electricity.

4.7 Methodological limitations

The analysis is descriptive, not causal. It identifies observed patterns but cannot establish why they occurred. January–November 2025 is treated as an incomplete year and compared with the same months in 2024 where year-on-year growth is reported.

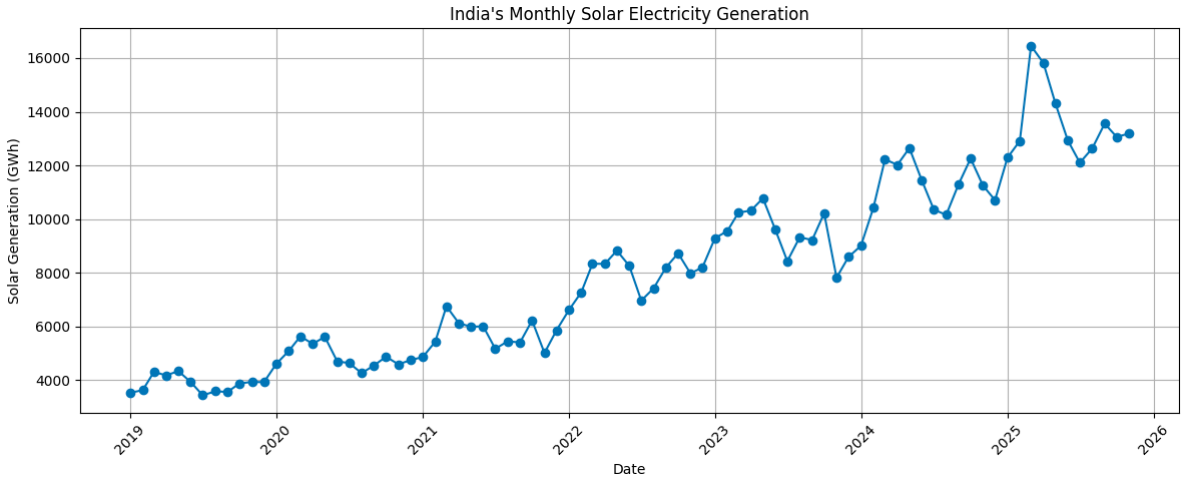
6. Analytical Questions

Analytical question	Method
How did solar and wind generation change?	Monthly and annual generation summaries (GWh).
How did solar and wind compare?	Combined-period totals and matched January–November growth.
Which states generated the most?	State-level ranking for 2024.
How did installed capacity change?	National capacity endpoint comparisons (MW).
What happened to renewable share?	National electricity-generation share series (%).
What happened to emissions?	Endpoint comparison of total emissions ($ktCO_2$) and intensity (gCO_2/kWh).

7. Solar and Wind Generation Analysis

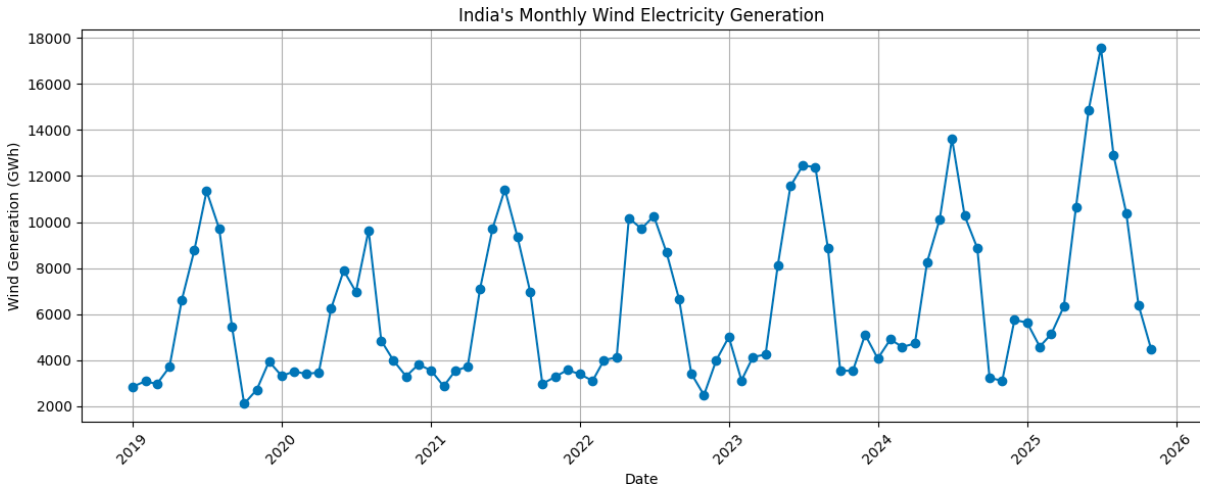
Across the full available period reported in the source document, solar generation totalled 664,860.01 GWh and wind generation totalled 524,529.16 GWh. Solar represented 55.9% and wind 44.1% of their combined solar-plus-wind total. These percentages are not shares of all Indian electricity generation.

Monthly solar generation trend



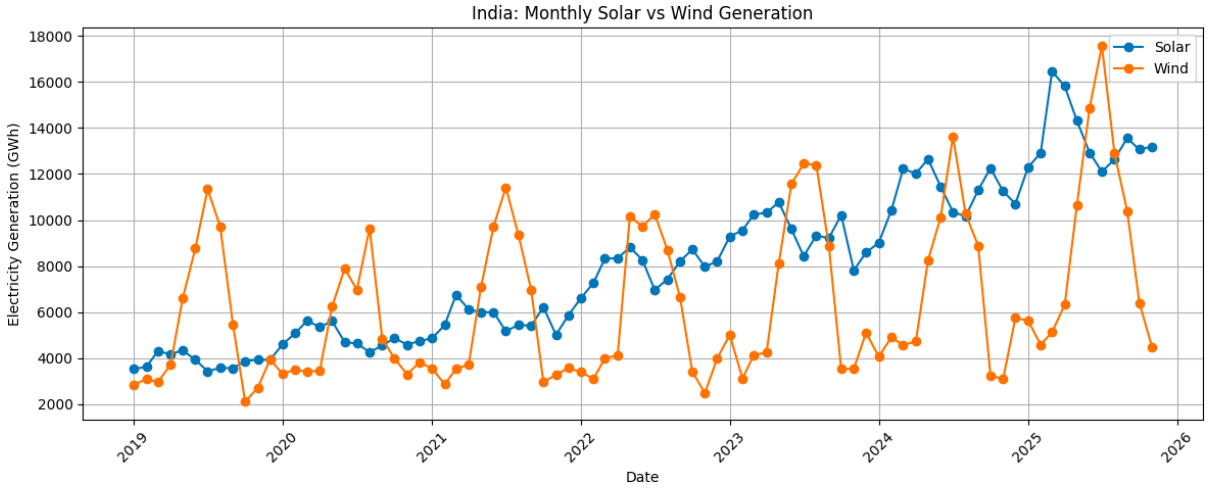
The solar chart shows an overall upward trend between January 2019 and November 2025, with month-to-month variation. The reported total across the available period is 664,860.01 GWh. The trend is consistent with a growing contribution from solar within the two technologies studied, but the chart alone does not establish the causes of growth.

Monthly wind generation trend



The wind chart also shows an overall upward trend, with visible seasonal variation. The report records annual wind generation of 63,314.28 GWh in 2019 and 81,544.17 GWh in 2024. January–November 2025 generation is reported as 98,997.47 GWh, 30.61% above the same months in 2024.

Solar and wind monthly generation comparison



The comparison shows different monthly patterns: solar trends upward over the period, while wind displays stronger seasonal fluctuations. The two sources may complement one another, although the plotted trends do not by themselves quantify reliability or prove a specific causal relationship.

Annual generation summary

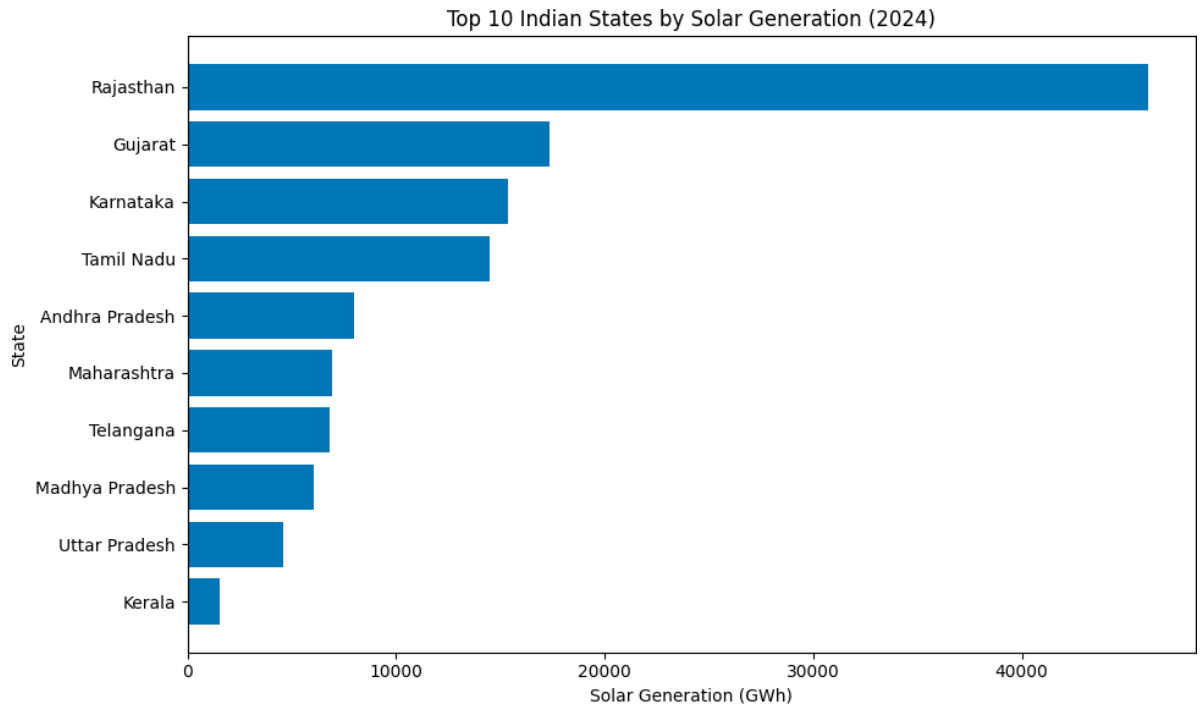
Period	Solar generation (GWh)	Wind generation (GWh)
2019	46,267.67	63,314.28
2020	58,683.51	60,427.42
2021	68,305.89	68,094.26
2022	95,160.31	70,045.17
2023	113,409.35	82,106.39
2024	133,795.05	81,544.17
2025 (Jan–Nov)	149,238.23	98,997.47

Matched-period comparison

Technology	Jan–Nov 2024 (GWh)	Jan–Nov 2025 (GWh)	Change
Solar	123,087.84	149,238.23	+21.25%
Wind	75,796.68	98,997.47	+30.61%

8. State-Level Analysis (2024)

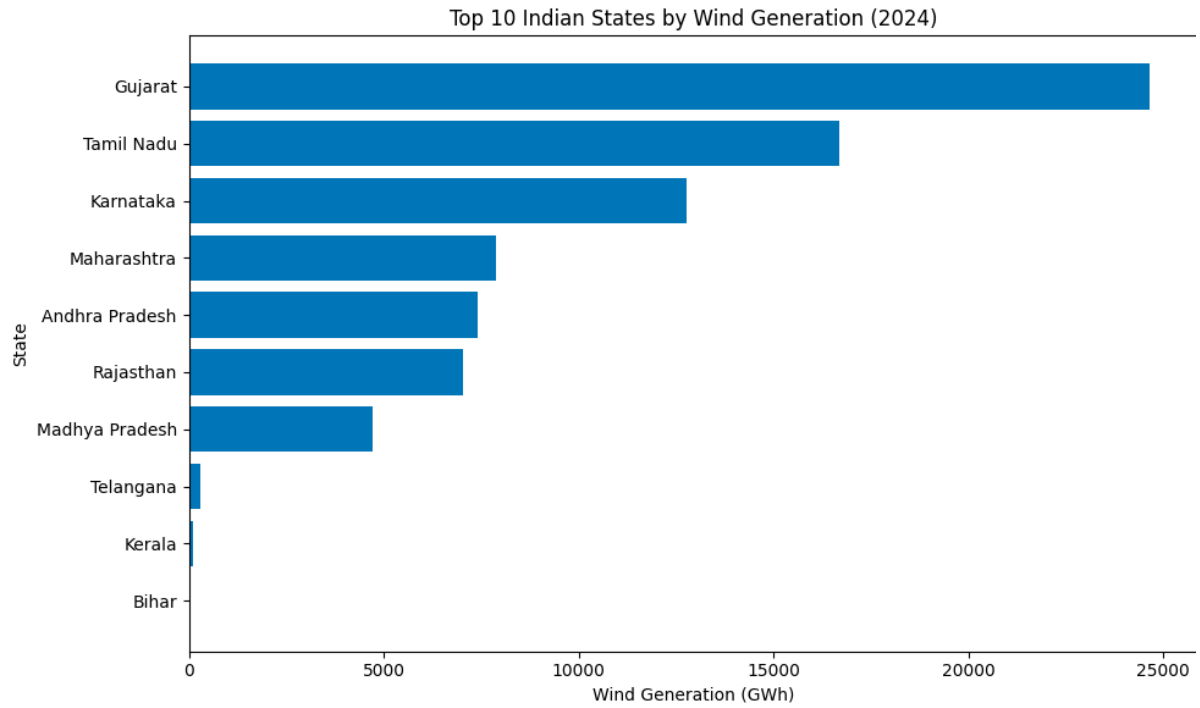
Top 10 solar-generating states



The report identifies Rajasthan as the leading solar-generating state in 2024 at 46,039.53 GWh, followed by Gujarat (17,374.37 GWh), Karnataka (15,397.94 GWh) and Tamil Nadu (14,476.31 GWh). Kerala is listed tenth at 1,561.62 GWh. Differences may reflect resource conditions, infrastructure and other factors, but those causes were not tested directly.

Rank	State	Solar generation (GWh)
1	Rajasthan	46,039.53
2	Gujarat	17,374.37
3	Karnataka	15,397.94
4	Tamil Nadu	14,476.31
5	Andhra Pradesh	7,978.50
6	Maharashtra	6,949.36
7	Telangana	6,838.33
8	Madhya Pradesh	6,032.95
9	Uttar Pradesh	4,604.39
10	Kerala	1,561.62

Top 10 wind-generating states

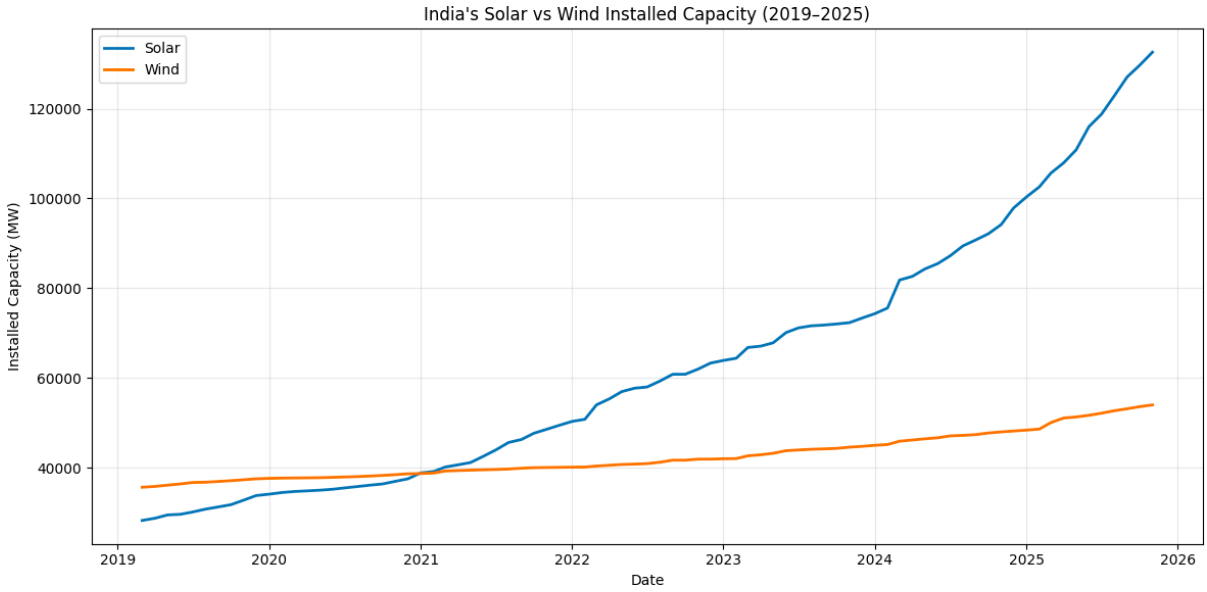


The report identifies Gujarat as the leading wind-generating state in 2024 at 24,644.66 GWh, followed by Tamil Nadu (16,685.71 GWh) and Karnataka (12,763.51 GWh).

Rank	State	Wind generation (GWh)
1	Gujarat	24,644.66
2	Tamil Nadu	16,685.71
3	Karnataka	12,763.51
4	Maharashtra	7,886.66
5	Andhra Pradesh	7,415.17
6	Rajasthan	7,032.20
7	Madhya Pradesh	4,704.10
8	Telangana	296.07
9	Kerala	116.07
10	Bihar	0.00

9. Installed Capacity

Solar and wind installed capacity trend



Solar capacity increased from 28,225.72 MW in March 2019 to 132,594.98 MW in November 2025, a reported increase of 369.77%. Wind capacity increased from 35,625.96 MW to 53,981.70 MW over the same endpoint dates, a reported increase of 51.52%. The different starting dates are preserved as reported.

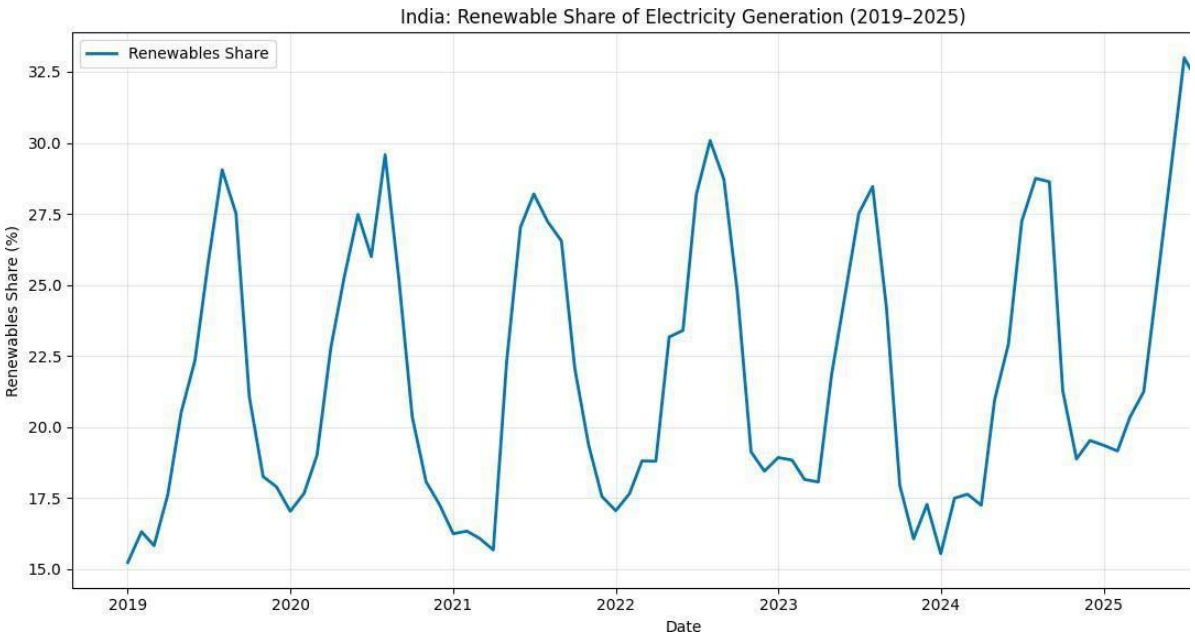
Technology	Capacity (MW), Nov 2025
Solar	132,594.98
Wind	53,981.70
Hydro	55,537.27
Bioenergy	11,610.18
Nuclear	8,780.00

Capacity measures installed power capability, not electricity delivered over a period. Generation (GWh) and capacity (MW) should therefore not be used interchangeably.

10. Renewable Electricity Share

The reported renewable share of national electricity generation rose from 15.24% in January 2019 to 23.01% in November 2025, an increase of 7.77 percentage points.

Renewable electricity share trend

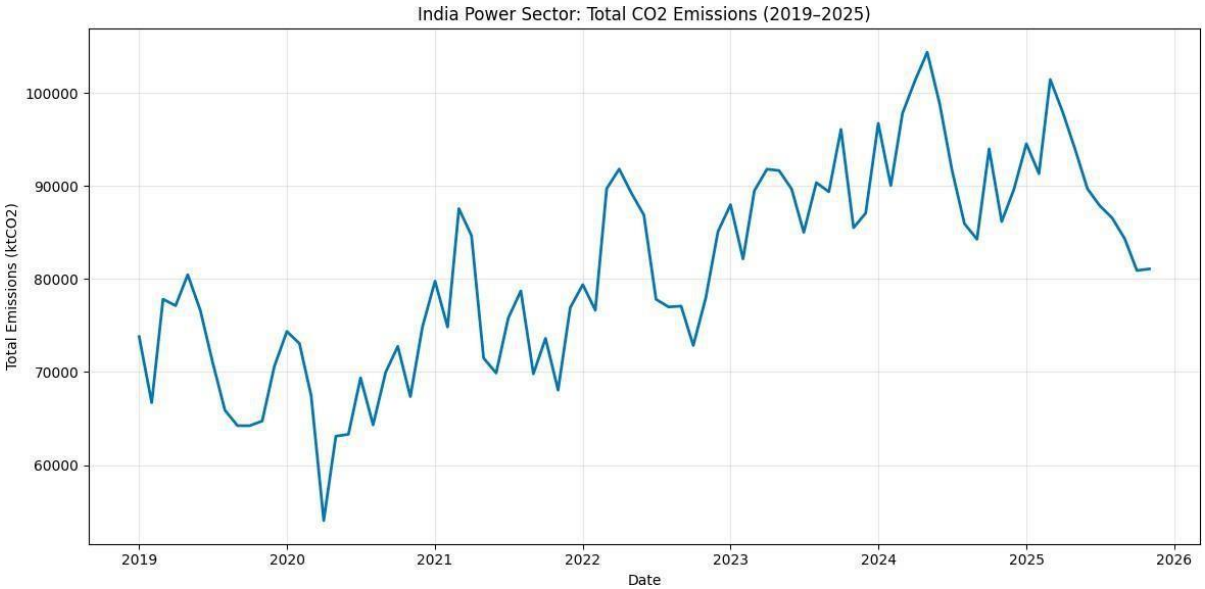


The series trends upward over the study period, although monthly values may vary with solar radiation, wind availability, hydropower output and total electricity generation. The endpoint comparison describes change in the reported share; it does not establish the causes of that change.

11. Emissions and Sustainability Findings

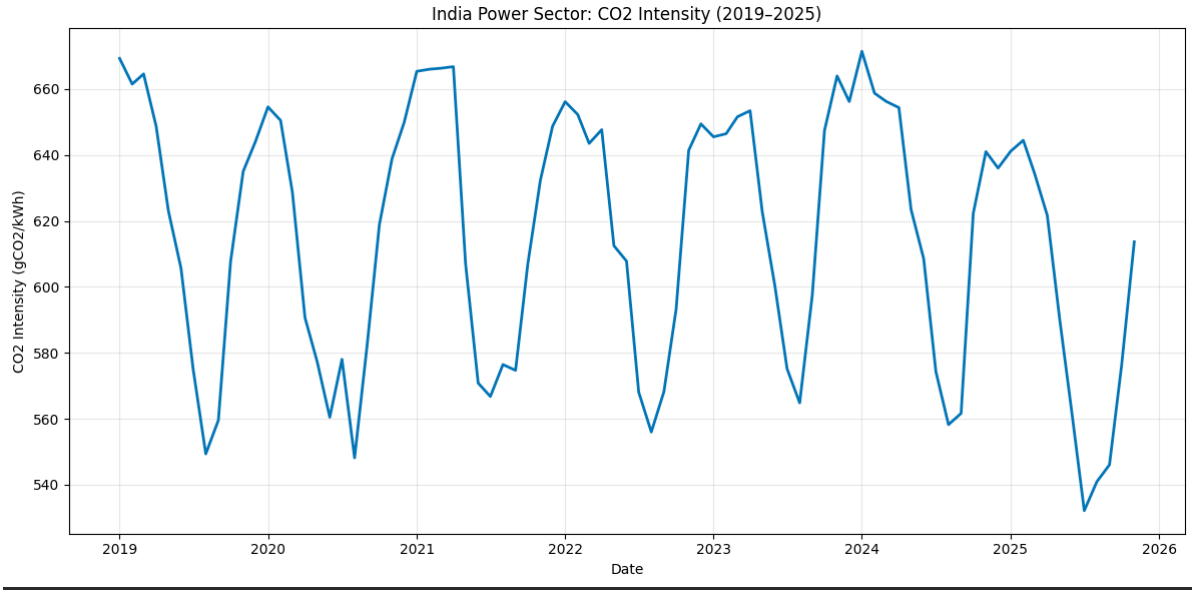
Total power-sector CO_2 emissions increased from 73,799.11 $ktCO_2$ in January 2019 to 81,078.26 $ktCO_2$ in November 2025, a reported increase of 9.86%.

Total power-sector CO_2 emissions trend



The endpoint values indicate higher total reported emissions at the end of the period. Changes in electricity demand, total generation and the generation mix could contribute, but this dataset and descriptive analysis do not establish the individual effects.

CO_2 intensity trend



CO_2 intensity declined from 669.27 gCO_2/kWh in January 2019 to 613.65 gCO_2/kWh in November 2025, a reported reduction of 8.31%. This means emissions per unit of electricity were lower at the later endpoint, according to the dataset.

Total emissions and emissions intensity can move in different directions. In this report, total emissions increased while intensity decreased. This does not prove that renewable expansion caused either change; total generation and other factors would need to be examined.

12. Recommendations

- Plan grid flexibility alongside renewable expansion. Transmission, storage and demand-response measures can help manage variable solar and wind output.
- Use state-specific planning. Tailor project development and grid investment to each state's resource profile, infrastructure and constraints.
- Track installed capacity and actual generation together. Capacity in MW measures capability; generation in GWh measures energy delivered over time.
- Monitor total emissions and intensity side by side. Use both $ktCO_2$ and gCO_2/kWh to avoid incomplete conclusions.
- Improve explanatory analysis by adding electricity demand, fossil-fuel mix, weather, storage and curtailment data.
- Validate data-quality flags, especially zero values, missing observations, units and reporting coverage, against the original source.

13. Limitations

- 2025 data ends in November and is not a full-year total.
- The original CSV, notebook, retrieval date and licence details were not included with the supplied PDF; those details should be confirmed from the original source.

- Endpoint comparisons do not describe every monthly change and may be affected by seasonality.
- Descriptive analysis cannot establish that renewable expansion caused changes in emissions or generation.
- State rankings can be affected by missing values, source definitions and reporting conventions.
- Capacity, generation, generation share, total emissions and emissions intensity are different measures and should not be interchanged.
- The reported zero for Bihar wind generation requires verification before drawing conclusions from the state ranking.

14. Conclusion

The analysis of renewable energy performance in India from January 2019 to November 2025 shows significant progress in the growth of renewable energy generation and installed capacity. The share of renewable energy in electricity generation increased from 15.24% in January 2019 to 23.01% in November 2025, indicating a growing contribution of renewable sources to India's electricity supply.

The study also shows substantial growth in solar and wind energy. Solar installed capacity increased from 28,225.72 MW in March 2019 to 132,594.98 MW in November 2025, while wind installed capacity increased from 35,625.96 MW to 53,981.70 MW. Furthermore, a comparison of January–November data shows that solar generation increased by 21.25% and wind generation by 30.61% in 2025 compared with the same period in 2024.

The analysis of carbon emissions reveals that total power-sector CO_2 emissions increased from 73,799.11 $ktCO_2$ in January 2019 to 81,078.26 $ktCO_2$ in November 2025. However, CO_2 intensity decreased from 669.27 to 613.65 gCO_2/kWh. This indicates that the emissions intensity of electricity generation declined, even though total emissions increased over the selected period. Therefore, the results show progress in renewable energy adoption but do not establish that renewable energy growth alone caused the reduction in CO_2 intensity.

In conclusion, renewable energy is playing an increasingly important role in India's electricity sector. Continued investment in solar and wind power, improvements in grid infrastructure, energy storage, and effective energy policies can support further progress towards a cleaner and more sustainable electricity system. This study provides a data-based overview of renewable energy trends in India during the selected period.